

TP2: proportional share CPU scheduler

Resources

- Resources for [assignment 1](#) remain useful
- Recommended reading: Xv6 book Sections 7.1–7.4

Tasks

1. We will implement the [Lottery Scheduler](#) as described in the OSTEP assignments.

Things to consider:

- Change the scheduler
 - Maintain the total number of tickets distributed to all running processes
 - Sleeping processes are not competing for CPU time and should not be considered
 - Consider if the variable holding the total number of tickets needs to be protected by a spinlock
 - Create a field to store the number of tickets each process holds (pstat)
 - Account for the new tickets whenever a process calls fork()
 - Inherit the number of tickets whenever a process calls exec()
 - Recover tickets whenever a process calls exit()
 - Change the scheduler to generate a random number between 1 and the number of tickets, select the appropriate process for execution
- Implement the system calls (settickets and getpinfo)

What to submit

You should upload a zip containing the complete version of your modified xv6 with the additional syscalls and lottery scheduler. You should also upload a documentation in PDF format with *at least* the following information:

- Discuss four modifications you made to the xv6 code; when appropriate, use the [git diff] command to compute the changes between the original xv6 code and your modified version.
 1. Show the diff and discuss the data structure you use to keep track of process tickets
 2. Show the diff and discuss the code you use to choose winning tickets (the random number generator)
 3. Discuss how you fill in the [pstat] structure
 4. Choose one more change you find relevant to discuss

- How you tested your scheduler
 - Discuss the [graph described at the end of the assignment description](#)